

ASSIGNMENT NO. 2

Title

Study and Demonstration of Git and GitHub Commands Using Terminal

1. Aim

To study Git and GitHub and demonstrate commonly used Git commands for creating, managing, tracking, and synchronizing a project repository using the terminal.

2. Introduction

Git is a distributed version control system (DVCS) used to track changes made to files during software development. It allows developers to maintain different versions of a project and view the history of changes.

Git works locally on a computer, so most Git operations can be performed without an internet connection. Developers can create commits, branches, and view the history of a project locally.

GitHub is a cloud-based platform that provides remote hosting for Git repositories. It allows developers to store their projects online, share source code, collaborate with other developers, and maintain project history.

Git and GitHub are commonly used together. Git manages the version history locally, while GitHub is used to store and synchronize the repository remotely.

3. Basic Git Concepts

Repository

A repository is a location where Git stores the files and complete version history of a project. A repository can be local or hosted remotely on platforms such as GitHub.

Working Directory

The working directory contains the actual project files on which the developer is currently working.

Staging Area

The staging area is an intermediate area where changes are placed before they are committed. The git add command is used to move changes to the staging area.

Commit

A commit is a saved version of the changes made to a project. Each commit has a unique identifier and a commit message describing the changes.

Branch

A branch is an independent line of development. Branches allow developers to work on new features or modifications without directly affecting the main project.

Remote Repository

A remote repository is a version of the project stored on a remote server such as GitHub. The local repository can be synchronized with the remote repository using commands such as git push.

4. Git Workflow

The basic Git workflow can be represented as:

Working Directory → Staging Area → Local Repository → Remote Repository

- Files are created or modified in the Working Directory.
- `git add .` moves the required changes to the Staging Area.
- `git commit` saves the changes in the Local Repository.
- `git push` uploads committed changes to the Remote Repository on GitHub.
- `git status` is used to check the current state of the repository.

5. Software Requirements

- Windows Operating System
- Git
- Visual Studio Code
- GitHub Account
- Internet Connection

6. Git and GitHub Commands

7. Check Git Version

Command:

```
git --version
```

Purpose: Checks whether Git is installed and displays the installed Git version.

2. Check Git Username

Command:

```
git config --global user.name
```

Purpose: Displays the username configured for Git commits.

3. Check Git Email

Command:

```
git config --global user.email
```

Purpose: Displays the email address configured for Git commits.

4. Check Repository Status

Command:

```
git status
```

Purpose: Shows the current state of the working directory and staging area.

5. Add Files to Staging Area

Command:

```
git add .
```

Purpose: Adds all modified and untracked files to the staging area.

6. Commit Changes

Command:

```
git commit -m "updated ASS1"
```

Purpose: Saves the staged changes as a new version in the local repository.

7. View Commit History

Command:

```
git log --oneline
```

Purpose: Displays the previous commits in a short format.

8. Check Branches

Command:

```
git branch
```

Purpose: Displays the branches available in the repository.

9. Create a New Branch

Command:

```
git branch assignment-test
```

Purpose: Creates a new branch named assignment-test.

10. Check Remote Repository

Command:

```
git remote -v
```

Purpose: Displays the URL of the remote repository connected to the local repository.

GitHub Repository:

https://github.com/aaradhyazankar/college_assignment.git

11. Switch to Main Branch

Command:

```
git switch main
```

Purpose: Switches from the current branch to the main branch.

12. Push Repository to GitHub

Command:

```
git push origin main
```

Purpose: Uploads the committed changes from the local main branch to the remote GitHub repository.

7. Conclusion

Git provides an effective method for version control and helps developers track changes made to a project. GitHub provides remote repository hosting and supports sharing and synchronization of projects.

Through this experiment, the basic Git workflow and commonly used Git and GitHub commands were successfully demonstrated using the terminal. The commands included checking the Git version, checking repository status, staging files, committing changes, viewing commit history, creating branches, checking the remote repository, switching branches, and pushing changes to GitHub.

Screenshots :

```
Microsoft Windows [Version 10.0.26200.9168]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\AARADHY\Desktop\project>git version  
git version 2.55.0.windows.3
```

```
C:\Users\AARADHY\Desktop\project>
```

```
C:\Users\AARADHY\Desktop\project>git config --global user.name  
Aaradhya Zankar
```

```
C:\Users\AARADHY\Desktop\project>git config --global user.email  
aaradhyazankar@gmail.com
```

```
C:\Users\AARADHY\Desktop\project>git status  
On branch main  
Your branch is up to date with 'origin/main'.
```

```
nothing to commit, working tree clean
```

```
C:\Users\AARADHY\Desktop\project>
```

```
C:\Users\AARADHY\Desktop\project>git status  
On branch main  
Your branch is up to date with 'origin/main'.
```

```
Changes not staged for commit:  
  (use "git add <file>..." to update what will be committed)  
  (use "git restore <file>..." to discard changes in working directory)  
    modified:   ASS1/index.html
```

```
no changes added to commit (use "git add" and/or "git commit -a")
```

```
C:\Users\AARADHY\Desktop\project>git add .
```

```
C:\Users\AARADHY\Desktop\project>git status  
On branch main  
Your branch is up to date with 'origin/main'.
```

```
Changes to be committed:  
  (use "git restore --staged <file>..." to unstage)  
    modified:   ASS1/index.html
```

```
C:\Users\AARADHY\Desktop\project>
```

```
C:\Users\AARADHY\Desktop\project>git commit -m "updated ASS1"
[main e37db06] updated ASS1
1 file changed, 1 insertion(+), 1 deletion(-)
```

```
C:\Users\AARADHY\Desktop\project>git log --oneline
e37db06 (HEAD -> main) updated ASS1
86128ec (origin/main) updated ASS1 Gym Membership Website
e82f507 Added ASS1 Gym Membership Website
a070455 first commit
```

```
C:\Users\AARADHY\Desktop\project>
```

```
C:\Users\AARADHY\Desktop\project>git branch
```

```
* main
```

```
C:\Users\AARADHY\Desktop\project>git branch assignment-test
```

```
C:\Users\AARADHY\Desktop\project>git branch
assignment-test
```

```
* main
```

```
C:\Users\AARADHY\Desktop\project>
```

```
C:\Users\AARADHY\Desktop\project>git remote -v
```

```
origin https://github.com/aaradhyazankar/college_assignment.git
(fetch)
```

```
origin https://github.com/aaradhyazankar/college_assignment.git
(push)
```

```
C:\Users\AARADHY\Desktop\project>
```

```
C:\Users\AARADHY\Desktop\project>git switch main
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
```

```
C:\Users\AARADHY\Desktop\project>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
```

```
nothing to commit, working tree clean
```

```
C:\Users\AARADHY\Desktop\project>
```